PES University, Bengaluru

(Established under Karnataka Act 16 of 2013)

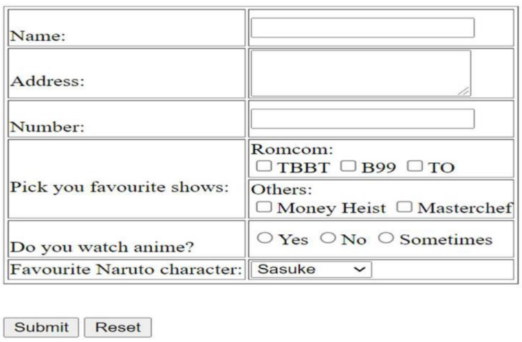
END SEMESTER ASSESSMENT (ESA) - Jan - May 2024

UE21CS242A - Web Technologies

Total Marks : 100.0

1.a. What is HTTP? Explain the structure of HTTP request message. (5.0 Marks)

1.b. Write a HTML code that outputs the following:



NOTE: The ‘Number’ field is a mandatory field, and takes in only a 10-digit number. Also, Submit and Reset buttons do their conventional tasks. (10.0 Marks)

1.c. The only way to apply CSS is by using external stylesheets”. Accept or reject this statement with suitable explanation. (5.0 Marks)

2.a. Write briefly about the two methods of geolocation object. (4.0 Marks)

2.b. A HTML page contains the following:

- A table exists with id “table1”
- A button “Add” with id “btn1”
- A div with id “display”

Add JavaScript to the webpage for the following specifications

• On clicking “Add” a new row with 2 cells are added to the table
• Populate each cell with a random number (between 1 and 200)
• If the cell contains an even number, mouse over should turn the cell green
• If the cell contains an odd number, mouse over should turn the cell red.
• If you click on a cell, the div will be populated with the cell content

Note: Write only vanilla JavaScript code. Assume that body tag has an onload=”init()” handler. (10.0 Marks)

2.c. Write jQuery code to perform the following:
1. For the last paragraph within the div with id “colortext”, set the color to green.
2. On moving the mouse over an h1 element, the text font size should be increased by 5 times
3. When the first li element with class name “liclass” on the page is clicked, it fades out in 2 seconds and then fades in 3 seconds (6.0 Marks)

3.a. Explain the following methods of component life cycle:
- componentDidMount
- componentDidUpdate
- componentWillUnmount (6.0 Marks)

3.b. Create a class component which will render a H1 and paragraph element apply different inline style for H1 and P element created using java script objects (key value pair). (6.0 Marks)

3.c. i) What would be the output of the following example? Justify your answer

```
var Test=(props)=>
{
return (
<div>
Hello World 1
</div>
<div>
Hello World 2
</div>
);
}
ReactDOM.render(<Test/>, mountNode)
```

ii) Which of the following is used to pass data to a component from outside in React.js? Justify your answer

a. SetState
b. Render with arguments
c. Props
d. PropTypes (6.0 Marks)

3.d. What is the significance of key property? (2.0 Marks)

4.a. Develop a basic server using the HTTP GET method that serves HTML files based on the provided pathname. If no pathname is provided, the server should default to serving 'index.html'. Additionally, implement error handling to display a 'File Not Found' message if the requested file is not found. (10.0 Marks)

4.b. Write JavaScript code using Node.js modules exclusively (without Express.js) to handle a GET request in the format http://server:8000/calc?op=add&op1=10&op2=20. The server should perform the specified arithmetic operation (addition, subtraction, multiplication, or division) on the two provided operands and return the result as the response. For instance, if the operation is 'add' and the operands are 10 and 20, the server should return the result of adding them, which is 30. (10.0 Marks)

5.a. What is RESTful API? Explain any 4 design specification/constraints of REST API. (6.0 Marks)

5.b. Write a program to upload a file to Node.js server using express file upload library. (6.0 Marks)

5.c. Develop a server-side JavaScript script to handle routing for GET and POST requests related to flight details. Flight details are stored in a MongoDB database in the following structure:

{from:”BLR”, to:”DEL”, dept:”12:25”, arrv:”14:25”, flnum:”6E-2428”}

The server script should support the following routes:
- GET /flights – return details of all flights
- GET /flights/:from/:to – return details of flight between specific airports
- POST /flights – save details of a flight and return a success or error message

(8.0 Marks)